

Abdelrahman Ahmed

Electrical Engineering Student

✉ bodebi639@gmail.com

in <https://www.linkedin.com/in/bitboody/>

☎ +20 10935791

📍 Cairo, Egypt

EDUCATION

MTI University

B.Sc Electrical Engineering

Sep 2024 - Present

WORK EXPERIENCE

Software Techies, Belgium (Remote)

Software Engineering Intern

Jun 2023 - Sep 2023

- Worked on optimizing a Node.js backend serving thousands of users, migrating it from JavaScript to TypeScript, and developing web scraping and data processing tools in GO to automate tasks and streamline workflows for engineers.

Software Engineering Intern

Jun 2022 - Sep 2022

- Collaborated with senior engineers on full-stack development projects, leveraging Node.js (with TypeScript), to design and implement scalable, efficient solutions.

PROJECTS

Odysseus (Open Source Maintainer)

- Serve as one of the maintainers of Odysseus, an open-source AI workspace platform with a large and active community.
- Review pull requests, mentor contributors, and help maintain code quality across the project.
- Develop new features, improve existing functionality, and resolve issues reported by the community.
- Write and maintain automated tests to ensure reliability and prevent regressions.

Radioactive decay simulator (Python)

- Developed a simulator to model radioactive decay, calculate half-lives, and predict gamma-ray emissions.
- Implemented mathematical models for radioactive decay and isotope behavior using exponential decay equations.
- Visualized simulation results to improve understanding of nuclear physics concepts.

6502 CPU Emulator (C)

- Developed a cycle-aware emulator for the MOS 6502 microprocessor.
- Implemented the complete official 6502 instruction set and addressing modes.
- Added support for interrupt handling and accurate processor status flag behavior.
- Modeled CPU execution with cycle-accurate timing.
- Verified emulator correctness using standard 6502 functional test ROMs.

Necromancer

- Developed a proof-of-concept remote administration tool (RAT) in Node.js for security research and experimentation.
- Implemented secure client-server communication and remote command execution.
- Explored networking, process management, and distributed system concepts in a controlled environment.